

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch : Mechanical Semester : V Subject Code & Name: BTAPE504D Automobile Engineering Max Marks: 60 Date: 08/02/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/ CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain the detailed classification of automobile.	CO1	6
B)	What are the main components of automobile? Explain any one of them.	CO1	6
C)	Explain different engine cylinder arrangements with neat sketches.	CO1	6
Q.2	Solve Any Two of the following.		12
A)	With the help of neat sketch describe steering linkage for rigid axle suspension.	CO2	6
B)	Define the steering geometry angles: Camber, king pin inclination, Caster, toe-in and toe-out on turns.	CO2	6
C)	Write down the functions and requirements of suspension system.	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Explain the working of single plate clutch with a neat diagram.	CO2	6
B)	What is the purpose of gear box? Explain the sliding mesh gear box.	CO3	6
C)	Write a note on stub axles.	CO2	6
Q.4	Solve Any Two of the following.		12
A)	Explain the working of hydraulic brake with a neat sketch.	CO2	6
B)	Draw a schematic diagram showing the layout of air brake system and explain functions of its main units.	CO2	6
C)	Write down the tyre-specification with meaning of each terms: 185 / 75 R14 82 S	CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Explain the construction and operation of lead acid battery with a neat sketch.	CO2	6
B)	Explain Standard Bendix drive with neat sketch.	CO3	6

C)	Write a short note on engine emissions.	CO5	6
	*** End ***		

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch: Mechanical Engineering Semester : Subject Code & Name: BTMEC506A Automobile Engineering Max Marks: 60 Date: Duration: 3 Hr.		
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Give detailed classification of automobile.	(L1/CO1)	6
B)	Draw transmission layout of an automobile.	(L2/CO2)	6
C)	Draw and explain tractive efforts and performance curves.	(L2/CO2)	6
Q.2	Solve Any Two of the following.		
A)	List different types of independent suspension systems and explain any one with neat sketch.	(L2/CO2)	6
B)	Draw and explain layout of steering system.	(L2/CO2)	6
C)	With neat sketch explain Davis steering mechanism.	(L2/CO2)	6
Q. 3	Solve Any One of the following.		
A)	With neat sketch explain the working of constant mesh gear box.	(L2/CO2)	6
B)	What is differential. Explain its working principle.	(L2/CO2)	6
C)	Write a note on fluid coupling and torque convertors.	(L2/CO2)	6
Q.4	Solve Any Two of the following.		
A)	Explain the working of disk brake and drum brake with neat sketch.	(L2/CO2)	6
B)	Differentiate between radial and cross-ply tyres.	(L2/CO2)	6
C)	List and explain different types of wheels.	(L2/CO2)	6
Q. 5	Solve Any One of the following.		
A)	With neat sketch explain the construction and working of ignition system.	(L2/CO2)	6
B)	Explain the construction and working principle of lead acid battery.	(L2/CO2)	6
C)	Write a note on emission control system and regulations.	(L2/CO2)	6
	*** End ***		

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2023 Course: B.Tech. Branch : Mechanical Engineering Semester : V Subject Code : BTAPE504D Subject Name: Automobile Engineering Max Marks: 60 Date: 17/08/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q.1	Solve Any Two of the following.		
A)	What are the basic components of an automobile system explain any four components in details?	1	6M
B)	What are the basic cylinder arrangements for automotive engines? Explain any three arrangements in detail?	1	6M
C)	What factors affect the tractive force requirement of a vehicle? Also explain vehicle performance curve?	1	6M
Q.2	Solve Any Two of the following.		
A)	Discuss in brief front axle? Explain any one type with neat sketch?	2	6M
B)	Define the following:- (1) Caster & Camber ? (2) toe-in & toe-out (3) king pin inclination.	2	6M
C)	Discuss in brief with neat sketch steering system layout.	2	6M
Q.3	Solve Any Two of the following.		
A)	What are the benefits of vehicle safety system? How many types of safety are there in car?	3	6M
B)	What are the different types of safety glass? Also mention properties and applications of safety glass?	3	6M
C)	Discuss in brief different types of car mirrors?	3	6M
Q.4	Solve Any Two of the following.		
A)	What are the basic requirement of wheels and tyres? Classify road wheels? Explain any one type with neat sketch.	4	6M
B)	Explain in brief wheel balancing? Also explain cross section of tubless tyre?	4	6M
C)	What are various factors affecting Tyre life and performance?	4	6M
Q.5	Solve Any One of the following.		
A)	Explain in brief Battery charging system? Also explain maintenance of lead acid batteries?	5	6M
B)	Explain in detail need of vehicle testing? Also explain vehicle test standards?	5	6M
C)	Discuss in brief emission control system and regulations.	5	6M
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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch: Mechanical & Allied Engineering

Semester: V

Subject Code & Name: BTAPE504D & Automobile Engineering

Max Marks: 60

Date:08-01-24

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

	(CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define automobile and mention the detail classification of it with example.	CO1	6
B) What is a chassis? Explain with the diagram front engine rear wheel drive chassis layout?	CO2	6
C) Enlist the main components of an automobile and describe them in detail?	CO2	6
Q.2 Solve Any Two of the following.		12
A) I) What are the Functions of steering system?	CO1	3
II) Differentiate clearly between the functions of spring and shock absorber of a suspension system.		3
B) Explain with the help of diagram the steering linkage of independent suspension system?	CO2	6
C) List out the common trouble experienced in automobile suspension system and suggest the remedies in each case.	CO4	6
Q. 3 Solve Any Two of the following.		12
A) Describe briefly with a neat sketch the construction and working of a single plate clutch.	CO3	6
B) Explain with a neat sketch the construction and working of a sliding mesh gearbox along with advantages and disadvantages?	CO2	6
C) Explain with a neat sketch the construction and working of a differential?	CO3	6
Q.4 Solve Any Two of the following.		12
A) What are the functions of brake? Classify it in detail with examples?	CO1	6
B) Draw a cross-section of automobile tyre and describe the basic constructional features of it?	CO2	6

C) With the help of neat sketch describe the operation of drum brake? CO3 6

Q. 5 Solve Any Two of the following. 12

A) Describe with neat sketch the construction and working of lead-acid battery used in automobile? CO3 6

B) Name the pollutants emitted by the IC engines in the atmosphere and mention its effects? CO5 6

C) Explain with a neat sketch the construction and working of magneto-coil ignition system in automobile? CO2 6

***** End *****